

Angles of elevation and depression



1 σ or $\angle CBD$

2 $x = 53.13^\circ$

3 $x = 37.23^\circ$

4 $\theta = 52.52^\circ$

5 $h = 19 \text{ m}$

6 $\theta = 41.81^\circ$

7 $x = 182$

8 $d = 229 \text{ m}$

9 $\theta = 23.9$

Applications

10
a $x = 455.84 \text{ m}$

b 425.84 m

11
a $h = 127.38 \text{ m}$

b 129.1 m

12 $d = 1022.58$

13 $h = 17 \text{ m}$

14 $d = 10.86$

15 $\theta = 45.97$

16
a $AB = 11.86 \text{ m}$ b $AC = 16.24 \text{ m}$ c $BC = 4.4 \text{ m}$

17

a $AC = 8663 \text{ m}$

b $BC = 6975 \text{ m}$



c 1688 m

18

a $x = 29.26^\circ$

b $y = 4.61^\circ$

c $z = 18.76^\circ$

d 100 cm

19

a $h = 9.37 \text{ m}$

b $\theta = 25^\circ$

20

a 61 m

b 1601 m

21

a $CD = 22.65 \text{ m}$

b $AB = 18.29 \text{ m}$